MILITARY SPECIFICATION

TELEGRAPH TERMINAL, TRANSISTORIZED VOICE FREQUENCY CARRIER AN/FCC-66( )

1. SCOPE

1.1 This specification covers objectives and technical requirements for multichannel frequency shift telegraph terminals capable of operating within normal radio and wire communication channels employed in Naval communications systems and facilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-Q-9858 - Quality Program Requirements
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and facilities

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-188 - Military Communication System technical Standards
MIL-STD-415 - Test Points and Test Facilities, Design Standard for
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470 - Maintainability Program Requirements for Systems and Equipment
MIL-STD-471 - Maintainability Demonstration
MIL-STD-781 - Reliability Tests Exponential Distribution
MIL-STD-785 - Requirements for Reliability Program
MIL-STD-1345 - Data, Measurement, In Support of Maintenance, Calibration and Repair of Electronic Equipment
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472 - Maintainability Prediction

PUBLICATIONS

MILITARY

NAVSHIPS-0967-312-8010 - Electronic Equipment Shipboard Maintainability Design Criteria Handbook or Designers of
NAVSHIPS-94501 - Reliability Design Handbook
NAVSHIPS-0969-125-0250 - Electronic Equipment Test for Prefered General Purpose

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEFENSE COMMUNICATIONS AGENCY

Circular 330-175-1

(Application for copies should be addressed to the Defense Communications Agency, General Offices Building 12, NSC, 8th and S. Courthouse Rd., Arl. Va. 22204.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 General requirements. The equipment shall be in accordance with the requirements of MIL-E-16400 to the extent specified herein.

3.2.1 Design objectives. The equipment shall be such as to attain the basic design objectives of MIL-E-16400 except that the Bureau of Naval Weapons reliability requirements and the magnetic characteristics paragraphs are not applicable.

3.2.2 Reliability.

3.2.2.1 Quantitative reliability requirements. The Voice frequency carrier telegraph (VFCT) terminal shall have a specified mean-time-between failure (MTBF) (θ_o) of 10,000 hours per channel. The MTBF for any piece part, modules, or subassembly whose failure would result in the failure of two or more telegraph channels shall be given by the formula MTBF 10,000 hours x channel failure (CF). CF is the number of channels that will fail if the piece-part, module, or subassembly fails.

3.2.2.2 Reliability program plan. The contractor shall prepare for approval a reliability program plan in accordance with the requirements of this specification and the requirements of MIL-STD-785. The reliability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program. The reliability program plan shall address the applicability or non-applicability of each paragraph of MIL-STD-785.

3.2.2.3 Reliability prediction. A design reliability prediction based on stress factors and part population shall be performed in accordance with MIL-HDBK-217. The prediction shall be based on the operational requirements of this specification. The reliability mathematical model and reliability block diagram shall be submitted with the reliability prediction report (see 6.1.2). Mechanical parts shall be included in the prediction. The source and applicability of the failure data used for the prediction shall be documented and justified prior to use. The results of the prediction shall be compared to the specified MTBF and where the predicted values do not achieve the requirements.

3.2.2.4 Reliability design guides. The contractor shall make maximum use of the design guides contained in NAVSHIPS 94501, Reliability Design Handbook.

3.2.2.5 Reliability letter reports. The contractor shall submit a letter report describing the progress and results of all reliability activity performed during the reporting period and the reliability activities planned for the next reporting period (see 6.1.2). Reports on the procedures, results and conclusions for the reliability activities of 3.2.2 and the reliability program plan shall be submitted with the reliability letter report (see 6.1.2).

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3.2.2.6 Reliability demonstration tests. The reliability testing shall be in accordance with 4.6.

3.2.3 Maintainability.

3.2.3.1 Quantitative maintainability requirement. The VFCT terminal shall have an equipment repair time (ERT) of no greater than 25 minutes.

3.2.3.2 Maintainability program plan. The contractor shall prepare for approval a maintainability program plan in accordance with the requirements of this procurement and the requirements of MIL-STD-470. The maintainability program plan shall address the applicability or non-applicability of each paragraph of MIL-STD-470. A preliminary maintainability program plan shall be submitted as part of the technical proposal (see 6.1.2).

3.2.3.3 Maintainability prediction. A maintainability prediction shall be performed using procedure II of MIL-HDBK-472. The contractor shall document and justify assumptions and the applicability of data used in the prediction. The prediction shall also include mechanical parts. The results of the prediction shall be compared to the required ERT and where the predicted values do not achieve the requirements the contractor shall accomplish such changes in design as are necessary to improve the predicted values sufficiently to achieve the requirements.

3.2.3.4 Maintainability design guides. The contractor shall make maximum use of the design guides contained in NAVSHIPS 0967-312-8010, Electronic Equipment Shipboard Maintainability Design Criteria Handbook or Designers of.

3.2.3.4.1 Circuit test points. Circuit test point accesses shall be provided which complies with the criteria outlined in MIL-STD-415. The following types of tests shall be considered for determining the test points: (1) signal quality, (2) circuit performance, (3) calibration, (4) system alignment, (5) installation and check-out, (6) line replaceable item (LRI) fault location. Test points for calibration shall be provided external to unit under test (UUT) except where it would affect the calibration. Test points shall be provided with proper interface fittings and matching circuits to connect the selected support test equipment.

3.2.3.4.2 Data, test point. Test point data shall be provided in accordance with MIL-STD-1345. Block diagrams for the complete system and for each LRI plus schematics shall be provided in sketch format at the same time the data is provided. Rationale for difficult test circuits shall also be provided with the test point data.

3.2.3.4.4 Support test instrumentation.

3.2.3.4.3.1 General purpose test equipment (GPTE). GPTE shall be selected from Electronic Equipment Test for Preferred General Purpose publication NAVSHIPS 0969-125-0250. The utilization of GPTE not on this list for support testing shall be requested in writing with justification consisting of parameters to be measured, techniques, accuracies, periodicity of tests, where tests are to be made, and so forth.

3.2.3.4.4 Test procedure verification. The test procedures of the Organizational and Maintenance (O&M) manuals for organizational, intermediate, and back-up support levels of maintenance shall be verified. The test equipment used to verify the procedures of the manual shall be the same as specified in the manual. Test equipment not specified in the manual may be used for the manual verification tests only if the contractor makes a written request and the command or agency concerned approves the request.

3.2.3.4.5 The Test verification shall proceed as follows:

3.2.3.4.5.1 Alignment. A technician will be given a complete set of boards, modules and components all of which are completely unaligned.

3.2.3.4.5.2 The technician will align each of the boards and verify that each module is operating properly.

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3.2.3.4.5.3 The technician will install the components of the system into the system and align the system to meet the requirements of this specification.

3.2.3.4.5.4 The technician will verify the procedures of the maintenance standards book and technical manual (TM).

3.2.3.4.5.5 The technician will calibrate the system as specified by the calibration procedure.

3.2.3.5 Maintainability letter report. The contractor shall submit a letter report describing the progress and results of all maintainability activity performed during the reporting period and the maintainability activites planned for the next reporting period (see 6.1.2).

3.2.3.6 Maintainability demonstration test. The maintainability requirements of the equipment shall be demonstrated in accordance with 4.7.

3.2.4 Human factor engineering. In order to realize the man-machine relationship from the standpoint of both operation and maintenance, the contractor shall comply with the human factor design standards of MIL-STD-1472 and the requirements specified herein.

3.2.4.1 Human engineering plan. A human engineering plan shall be submitted in accordance with the Human Engineering Plan paragraph of the appendix of MIL-H-46855 as part of the contractor's original bid and revised and resubmitted after contract award (see 6.1.2). The plan shall include, as a minimum, provisions for the compliance with the requirements of the Failure Analysis paragraph, the "Drawing Approval" paragraph of the appendix of MIL-H-46855 and the Controls, Indicators, and Panel Layout paragraph of MIL-E-16400.

3.2.4.2 Guidance meeting. The contractor shall arrange a meeting with appropriate representatives of the command or agency concerned for the purpose of obtaining guidance on complying with the human factor requirements of this specification. The meeting shall be directed toward achieving mutual understanding of information contained in the contractor's proposal and the requirements imposed by this specification.

3.2.4.3 Equipment requiring human engineering review. A list of equipment requiring human engineering design detail for operation and maintenance shall be prepared as required by MIL-H-46855 and submitted to the command or agency concerned. This list shall be initially submitted after contract award and updated and resubmitted (see 6.1.2), separating the equipments into operator and maintainer man-machine interfaces. It shall describe, as a minimum, which front panel configurations and equipment interior views should be reviewed for conformance to the human engineering design criteria of this specification.

3.2.4.4 Auditory alarms. The requirements of MIL-STD-1472 for auditory displays shall be observed in the design of a summary audible alarm system. This system shall respond to blown fuses as required by 3.17.4.1.3 and to other alarms indicated by alarm lamps on the front panels. The audio warning device shall have an indicating lamp which is not extinguished by the disabling switch, and it shall have a disabling switch with automatic reset circuitry.

3.2.4.5 Human engineering design verification. A human engineering verification test plan to test the equipment and system of this specification shall be prepared in accordance with the requirements of the "Failure Analysis" paragraph and the "Human Engineering Design Verification - Test and Evaluation" paragraph of MIL-H-46855. This plan, to include human factor and maintenance checklists to be supplied by the Human Factors code of the command or agency concerned, is to be implemented as part of the equipment and system test associated with the contractor's First article or prototype acceptance test program.

3.2.4.6 Human engineering progress reports. Progress reporting on human engineering work shall conform to the requirements listed in the appendix of MIL-H-46855 except that the progress reports shall be submitted at least twice to the command or agency concerned. The first report is to be submitted prior to the human engineering design verification testing of 3.2.4.5 and the second report after testing (see 6.1.2).

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3.2.4.7 Change requests. In the event the contractor wishes to make a change in the approved human engineering plan, 3.2.4.1, the approved panel drawings, 3.1.4 or the approved human engineering verification plan 3.2.4.5 a change request shall be submitted to the command or agency concerned for approval. This request, shall be submitted prior to the effective date of the change, and shall state what the change is, why it is required, and its effect on the system, equipment or facilities (see 6.1.2).

3.3 Definitions. The definitions of terms used in this specification shall be in accordance with MIL-E-16400.

3.4 Modular construction for equipment. Modular construction shall be in accordance with MIL-E-16400.

3.5 Parts. Parts shall be in accordance with MIL-E-16400.

3.6 Material. Material shall be in accordance with MIL-E-16400 except that radioactive material shall not be used in any form and the following phrase and paragraphs do not apply:

(a) The magnetic permeability phrase under metals.
(b) Equipment subject to sea water immersion.
(c) Magnetic materials.

3.7 Coatings and treatment. Coatings and treatment shall be in accordance with MIL-E-16400.

3.8 Safety. In order to realize safe operation and maintenance of the equipment of 1.1 from the standpoint of safety for personnel and equipment, the contractor shall comply with the safety design standards in MIL-E-16400. In addition, the following requirements shall be met.

3.8.1 Voltage below 30 volts (rms or dc) shall be used whenever possible.

3.8.2 All access doors which may be opened to expose voltages in excess of 30 volts during operation and maintenance shall be clearly marked "DANGER-VOLTAGE DANGEROUS TO LIFE." The letters on the marking shall be gothic capitals, clearly legible, color white with red background.

3.8.3 The voltage terminals for all voltages in excess of 30 volts which are not deactivated by an interlock circuit when an access door is opened shall be protected by barriers which are clearly marked "CAUTION - HIGH VOLTAGE." The letters on the marking shall be gothic capitals, clearly legible, color black with yellow background.

3.8.4 The interlock over-ride switch shall be located on the front of the equipment. It shall be possible when the switch is in the over-ride position to open the rear access door without interrupting the power to the unit. In the non-over-ride position whenever the rear door is open the power to the unit shall be interrupted.

3.9 Environmental conditions. The equipment shall meet the requirements of MIL-E-16400 for temperature (Class 4), and for humidity.

3.10. Electrical design. Electrical design shall be in accordance with MIL-E-16400 except that the preferred circuits and the internal supply voltages paragraphs do not apply.

3.10.1 Blown fuse indicators and an alarm shall be required.

3.10.2 The equipment shall operate with primary power line voltages of 115 volts or 230 volts ±10 percent.

3.10.3 The equipment shall operate with primary power line frequencies of 47 to 440 hertz (Hz).

3.11 Maintenance design. Maintenance design shall be in accordance with MIL-E-16400.

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3.11.1 One double receptable with ground terminals shall be provided on the front of the cabinet located near the bottom and shall be entirely separate from ac power to the VFCT terminal.

3.12 Mechanical design. Mechanical design shall be in accordance with MIL-E-16400 except that the following do not apply:

(a) Shock, vibration, inclination and noise
(b) Enclosures
(c) Size
(d) Handling

3.13 Thermal design. Thermal design shall be in accordance with MIL-E-16400 except that the heat removal method shall be as specified for equipment which is not intended for shipboard use. Each terminal shall contain an exhaust fan which shall vent out of the top of the cabinet.

3.14 Controls, indicators and panel layout. Controls, indicators and panel layout shall be in accordance with MIL-E-16400, except that time meters shall not be required.

3.15 Supplementary requirements. Supplementary requirements shall be in accordance with MIL-E-16400 except that the Bureau of Naval Weapons contracts paragraph does not apply.

3.16 Electromagnetic interference emission, susceptibility and systems self-compatibility. The equipment shall conform to the requirements of class 1C of MIL-STD-461. The failure definition for susceptibility purposes shall be as defined in 4.5.4. The EMI electromagnetic interference control plan shall be submitted for approval (see 6.1.2). The EMI/EMC test plan shall be submitted prior to commencement of the electromagnetic characteristics EMC test (see 6.1.2). The EMC test report shall be submitted after completion of the test (see 6.1.2).

3.17 Functional requirements.

3.17.1 Telegraph terminal. The telegraph terminal shall be solid state. Active or rectifying electrical elements shall be semiconductor devices. Electromechanical relays shall not be used. Vacuum tubes shall not be permitted. The terminal shall be capable of operating over a communications channel with the AN/FCC-66. The input and output devices shall conform to 3.17.3.1 and 3.17.3.2.

3.17.1.1 Terminals shall be completely wired, harnessed and factory adjusted so as to be operational in the configuration specified after initial on-site installation and adjustments for interface signal conditions.

3.17.2 Each telegraph terminal shall be based on modular building block techniques. The 16 channel diversity terminal shall be capable of being configured in the arrangement described herein.

3.17.2.1 Configuration.

(a) 16 channel full duplex, dual and quad diversity operating with a bandwidth of 340 to 3060 Hz.

3.17.2.1.1 Dual diversity, or quadruple diversity.

3.17.2.2.1 Dual diversity as used in this specification shall depend on the nature of the transmission media. Dual diversity shall be accomplished either by space diversity or frequency diversity operation of the transmission media.

3.17.2.2.2 Frequency diversity as applied to the device for this specification is defined as the transmission if identical information on two different channels within a single 3 kilohertz (kHz) band. This function is defined as in-band diversity.

3.17.2.2.3 Quadruple diversity is the combination of diversity operation on the transmission media and in-band diversity.

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3.17.3 The telegraph terminal shall receive from and transmit to telegraph loops as follows:

3.17.3.1 The transmit section of each telegraph terminal shall accept input telegraph signals in any of the following modes:

Operating Mode Options	Operating Voltage	Operating Current
POLAR, Low Level (positive and negative battery)	±6 Volts (±1V)	±1.0 mA
POLAR, High Level (positive and negative battery)	±60 Volts (±1.2V)	20 mA (±2 mA)
NEUTRAL, (positive or negative battery)	130 Volts (±2.6V)	60 mA (±1.5 mA)

3.17.3.1.1 The input characteristics shall be as specified in the DC Digital Transmission Reference Systems of MIL-STD-188 for high performance facilities except that the terminal shall be required to accept input signals containing up to 40 percent total peak telegraphic distortion. It is not required that output signals from the transmit section contain less distortion than the corresponding input signals.

3.17.3.1.2 The mode shall be independent for each telegraph loop and selectable by strapping appropriate terminals which are readily accessible.

3.17.3.1.3 The safety factor for high level modes shall be at least ±2.5; that is when a channel is strapped for x milliamperes (mA) and y volts no damage shall result if the input signal is ±2.5 x mA and ±2.5 y volts.

3.17.3.1.4 The safety factor for low level polar shall be ±6 for both current and voltage.

3.17.3.2 The receive section of each telegraph terminal shall generate output telegraph signals in any of the following modes:

(a) 60 mA, 130 ±2.6 (high level neutral)
(b) 20 mA, 60 ±1.2 (high level polar)
(c) 1 mA, 6 ±1 (low level polar)

3.17.3.2.1 The output characteristics shall be as specified in the DC Digital Transmission Reference Systems of MIL-STD-188 for high performance facilities. Suitable protection shall be provided for each output telegraph loop.

3.17.3.2.2 The mode shall be independent for each loop and selectable on a strapping basis.

3.17.4 The telegraph terminal shall receive from and transmit to communications channels whose transmission media may be metallic circuits, microwave or high frequency (hf) radio.

3.17.4.1 When a terminal is in its normal operating environment and performing within the requirements of this specification, the performance of the terminal shall not be degraded beyond the limits of this specification if any normal operational adjustments are made to the terminal within the requirements of this specification, such as signal level, polarity or modulation rate.

3.17.4.1.1 It shall be possible to operate the 16 channel, full duplex, diversity terminal in the following subsystem basis;

(a) 4 separate subsystems of 4 channels each
(b) 2 separate subsystems 4 channels each and one separate subsystem of 8 channels
(c) 2 separate subsystems of 8 channels

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(d) 1 separate subsystem of 12 channels and 1 separate subsystem of 4 channels
(e) 1 system of 16 channels

3.17.4.1.1.1 The various subsystem operating shall be accomplished by setting switches on the front of the terminal. In any of these subsystems the terminal shall operate within the requirements of this specification without any adjustment or alignment. There shall be independent switching arrangements for the transmit section(s) and the receive section(s) for the diversity terminal.

3.17.4.1.1.2 Each configuration in 3.17.4.1.1 shall interface with a land-line or microwave link. The interface provided in any mode of operation shall be: 600 ohm balanced with center tap available; minimum return loss of 26 dB over the frequency range 370 to 3040 Hz; 40 dB longitudinal current suppression.

3.17.4.1.1.3 The single line drawings, Figures 1 and 2, depict the output circuitry of the keyers and the input circuitry of the converters of each diversity telegraph terminal and may be used to interpret the requirements specified in 3.17.4.1.1.

3.17.4.1.1.4 Terminals shall be wired and harnessed and otherwise furnished in order to be operable under the subsystem concept specified in 3.17.4.1.1 except that the number of amplifiers to be furnished with each terminal shall operate as follows:

(a) 16 channel full duplex diversity (2 on line, 1 spare)

3.17.4.1.1.4.1 It shall be possible on a strapping basis of inserting the spare line amplifier in the composite output lines of the terminal.

3.17.4.1.2 Blown fuse indicators shall be required for each module. In the event of a power supply failure that is common to more than 2 modules, a fuse indicator shall be activated. Each fuse indicator shall be capable of activating a common audible alarm when a failure occurs. Fuse indication shall be located on the front panel.

3.17.4.2 When the composite output signal of the transmit section, in any configuration is connected on a back to back basis to: (1) the first diversity receive section only; (2) the second diversity receive section only; (3) the first and second diversity receive sections, the following requirements shall be met.

3.17.4.2.1 When all output channels are continuously keyed with 90 baud random signals (the input to any channel is completely independent of the input to any other channel) with less than 1 percent distortion in any of the modes specified in 3.17.3, the telegraph signals appearing at the output of each receive channel shall not exceed 5 percent total peak telegraphic distortion when all keyers are adjusted so that the input signals to each converter are: (1) All at the level of -40dBm; (2) All at the level of -10dBm; (3) All at any level in the range of -40dBm to -10dBm. Under the same conditions the total peak telegraphic distortion shall not exceed 10 percent when input signals to the converters are in the range of -50dBm to -40dBm and -10dBm to 0dBm. When any channel is idled out, the adjacent channel interference measured on the loop side of the idle channel filters shall not exceed -60dBm.

3.17.4.2.1.1 When the terminal is operating with the input signals to each converter at -20.7dBm, if any one of the transmit channels is attenuated -30dB, the output telegraph signal from the corresponding receive channel shall not have more than 40 percent total peak telegraphic distortion.

3.17.4.2.1.2 When the transmit section is connected to the first and second diversity receive sections, either receive section shall be capable of accepting a -50dB fade of the composite signal without increasing the total peak telegraphic distortion beyond the specified limits.

3.17.4.2.2 When the input channels are keyed with the same 90 baud random signals in any of the modes specified in 3.17.3 the differential delay between any two output channels shall be adjustable within 500 microseconds.

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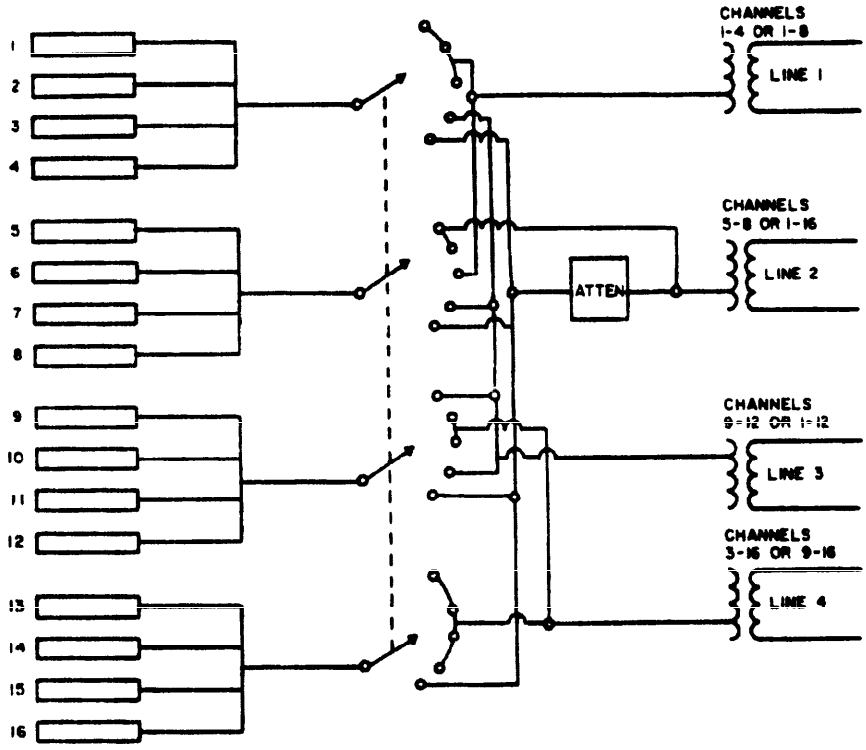
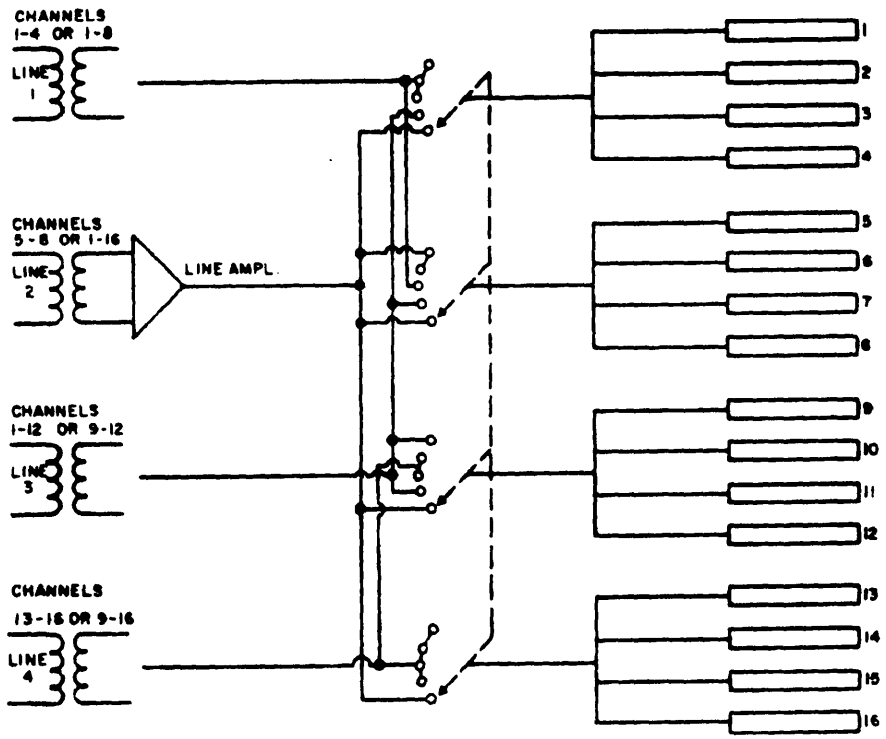


Figure 1 - Output Circuitry of Keyers.

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Note: For diversity configurations a second receiving circuitry identical to above shall be provided.

Figure 2 - Input Circuitry of Converters.

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3.17.5 The necessary parts that comprise the following shall be housed in one standard 19 inch cabinet (similar to a CY-597) with a maximum over-all cabinet height of 80 inches.

(a) 16 channel, fullduplex - dual and quad diversity
(b) Test panel

3.17.5.1 The VFCT terminal shall be completely installable and maintainable from the front of the cabinet.

3.17.5.2 Rear access door shall be provided on all cabinets.

3.17.5.3 The VFCT terminals shall be comprised of items included in 3.17.5.3.1 thru 3.17.5.3.7.6. The number of items shall be in accordance with the configuration specified. The physical configurations of the parts or modules need not coincide precisely with the description, provided that all other requirements are satisfied.

3.17.5.3.1 Keyers. Unless otherwise specified, keyers shall be narrow band as follows: One frequency shift keyer for each transmit channel shall be required for all configurations. A keyer shall accept an input signal at any modulation rate up to 90 baud at any of the modes specified and convert it to an audio signal as specified in Table I.

Table I — Channel allocations

Channel Number	Mark Frequency (Hz)	Center Frequency (Hz)	Space Frequency (Hz)	Sending Signal Sense (Non-diversity)
1	382.5	425	467.5	-
2	552.5	595	637.5	-
3	722.5	765	807.5	-
4	892.5	935	977.5	-
5	1062.5	1105	1147.5	-
6	1232.5	1275	1317.5	-
7	1402.5	1445	1487.5	-
8	1572.5	1615	1657.5	-
9	1742.5	1785	1827.5	-
10	1912.5	1955	1997.5	-
11	2082.5	2125	2167.5	-
12	2252.5	2295	2337.5	-
13	2422.5	2465	2507.5	-
14	2592.5	2635	2677.5	-
15	2762.5	2805	2847.5	-
16	2832.5	2975	3017.5	-

3.17.5.3.1.1 Adjacent channel separation shall be 170 Hz.

3.17.5.3.1.2 The frequency deviation from the center frequency shall be ±42.5 Hz.

3.17.5.3.1.3 Provision shall be made for inverting the output signal mark-space frequency assignment with respect to the input signal for any keyer as specified in 3.17.5.3.1. This shall be accomplished by setting a switch or strap on the keyer that is readily accessible but not located on the front of the module.

3.17.5.3.1.4 The output level shall be continuously adjustable over a minimum range of 0 dBm to -30 dBm with a single control.

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3.17.5.3.1.5 The mark and space output signal levels for any one keyer shall be within 1.5 dBm of each other.

3.17.5.3.1.6 The harmonic content output of any keyer shall be 60 dB below the out signal.

3.17.5.3.1.7 Keyers shall have the same physical dimensions and mounting requirements. The units shall be identical for any operating frequency and frequency-shift except for the modular plug-in subassemblies which determine the operating frequency and shift. Each keyer shall provide for mounting a channel designator nameplate on the front. The nameplate shall be furnished with the frequency and frequency-shift elements and shall designate the mark, space and center frequencies.

3.17.5.3.1.8 Readily accessible strapping terminals shall be provided on each keyer in order to select the mode of input signal as specified in 3.17.3.1.

3.17.5.3.1.9 Test monitor jacks for the essential test points shall be provided on the front panel of each keyer. The minimum acceptable front panel monitor jacks shall be as follows:

(a) Input telegraph signal
(b) Keyer audio output signal
(c) Input signal to bandpass filter

3.17.5.3.1.10 Each keyer shall be capable of transmitting continuous mark, continuous space, of accepting keying from the external signal line or disabling the keyer output without removing the power from the unit. This shall be accomplished by setting a switch located on the front panel of the module.

3.17.5.3.1.11 No adjustments shall be required except for setting the output level frequency and key-over point. The frequency shall be adjustable or crystal controlled to within ±0.1 Hz from the specified frequency. The frequency and sensitivity adjustments, if required, shall be readily accessible but not located on the front panel of the keyer. The output level adjustment shall be located on the front panel of the keyer. The sensitivity of the keyer shall be as follows for the different operating modes:

Mode	Operating Differential	Centered at
High Level Polar, 2mA	2 to 4 mA	0 mA
Neutral, 60mA	2 to 4 mA	30 mA

In the case of the 6 Volt polar signal, a marking current not in excess of 100 microamperes shall cause the receive device to correctly assume the marking (one) state, while a spacing current of the same magnitude shall cause the receive device to correctly assume the spacing (zero) state.

3.17.5.3.1.12 The input shall be electrically floating so that either or neither input terminal may be grounded at the option of the user by strapping appropriate terminals which are readily accessible.

3.17.5.3.1.13 The frequency characteristics of any channel at the mark and space frequencies shall be as follows.

3.17.5.3.1.13.1 The mark and space transmitted frequencies shall be with ±0.1 Hz of the specified values with stability such that the deviation from the transmitted frequencies shall be no greater than ±0.5 Hz over a six month period. Capability shall be provided to permit adjustment of the mark and space frequencies to within ±0.1 over a minimum period of 5 years.

3.17.5.3.1.14 The jitter shall be less than 2 percent for any frequency.

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3.17.5.3.2 Convertor/loop keyer. Convertors shall be narrow band. Two frequency shift convertor and one loop keyer for each receive channel shall be required for diversity configuration. The frequency assignment shall be identical to that of the keyers as specified in 3.17.5.3.1. Each convertor/loop keyer shall select the assigned channel frequency and discriminate against the unwanted channels. The selected channel frequency shall be converted to telegraph signals as specified in 3.17.3.2. The power requirements (loop battery) for all specified modes of telegraph signals shall be provided wholly by each loop keyer and AC shall be directly from the primary AC power line. Provision shall be made for use of an optional external battery (GFE) for telegraph loops. The DC voltage regulation, ripple limits and other parameters are specified in 3.2.3.7 of DCA circular 330-175-1. Each loop keyer battery shall be capable of providing 60 milliamperes into a varying output load of 50 to 1800 ohms.

3.17.5.3.2.1 Provision shall be made for inverting the mark and space sense of the output signal with respect to the input signal for any convertor. This shall be accomplished by setting a switch on the convertor that is readily accessible but not located on the front of the module.

3.17.5.3.2.2 The convertor/loop keyer shall perform within the requirements of this specification without readjustment when receiving a signal from the corresponding transmitter at any level in the range from -50 dBm to 0 dBm or when the signal is varied within this range at a rate encountered in fast fading hf radio.

3.17.5.3.2.3 The bias distortion shall be continuously adjustable over a minimum range of ±10 percent at any rate between 45 and 90 baud. When the input signal is anywhere within its rated limits and contains less than 1 percent bias distortion the output from the convertor shall be adjustable in order to produce less than 1 percent bias distortion.

3.17.5.3.2.3.1 When the frequency of the input signal (at any rate up to 90 baud) varies ±5 Hz from its assigned values, the total peak telegraphic distortion from the convertor/loop keyer shall not exceed 15 percent.

3.17.5.3.2.4 The time delay of the output telegraph signal shall be continuously adjustable over a minimum range of zero to 10 milliseconds.

3.17.5.3.2.5 Convertors and loop keyers shall have the same physical dimension and mounting requirements. The units shall be identical for any operating frequency and frequency shift except for the modular plug-in-subassemblies which determine the operating frequency and shift. Each convertor shall provide for mounting a channel designator nameplate on the front. The nameplate shall be furnished with the frequency and frequency shift elements and shall designate the mark, space and center frequencies.

3.17.5.3.2.6 Reading accessible strapping terminals shall be provided on each loop keyer in order to select the mode of output signal as specified in 3.17.3.2.

3.17.5.3.2.7 Test monitor jacks for all essential test points shall be provided on the front panel of each convertor/loop keyer. The minimum acceptable front panel jacks shall be as follows:

(a) Audio signal on loop side of telegraph channel bandpass filter
(b) Output signal from discriminator
(c) Output telegraph signal

3.17.5.3.2.8 The output shall be electrically floating so that either or neither output terminal may be grounded at the option of the user by strapping appropriate terminals which are readily accessible.

3.17.5.3.3 Diversity combiners. One combiner for each pair of receive convertors shall be required for the diversity configuration. Convertor twinning for frequency diversity shall be as shown in 3.17.5.3.4. Provisions shall be made for selecting the mode of operation by a switch mounted on the front panel of the combiner module marked as follows (a) dual diversity, and (b) quadruple diversity. It shall not be necessary to make any adjustments when the mode is changed.

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3.17.5.3.4 Keyer twinning panels for local and remote shall be furnished for each diversity terminal. Keyers shall be twinned on the adjacent numerical channels shown herein, for example 1 and 2, 3 and 4. When twinning is employed, the even numbered input channel shall be keyed simultaneously with its associated odd numbered channel, and even number channel output shall be combined with the associated odd channel in comparator circuits to produce a single output from the odd channel. The DC telegraph keying loops on even channel inputs and output shall be open circuited.

Channel number	Center frequency diversity	Sending signal sense diversity
1	1785	-
2	425	-
3	1955	-
4	595	-
5	2125	-
6	765	-
7	2295	-
8	935	-
9	2465	-
10	1105	-
11	2635	-
12	1275	-
13	2805	-
14	1445	-
15	2975	-
16	1615	-

3.17.5.3.5 Amplifier.

3.17.5.3.5.1 Frequency response shall be flat ±2 dB from 300 Hz to 6000 Hz.

3.17.5.3.5.2 The input and output impedance shall be 600 ohms balanced, with a minimum return loss of 26 dB over the frequency range 300 to 600 Hz. The longitudinal current suppression shall be at least 40 dB below the input or output signal Transformer coupling shall be required.

3.17.5.3.5.3 The gain shall be continuously adjustable, from 0 dB to 35 dB when the input level is between -50 dBm and 0 dBm the maximum value of any harmonic generated by the amplifier shall be at 40 dB below the output level for any level up to +10 dBm. The hum and noise level shall be at least 40 dB down from 0 dBm output.

3.17.5.3.5.4 Test monitor jacks for all essential test points shall be provided on the front panel of each amplifier. The minimum acceptable front panel jacks shall be as follows:

(a) Input signal
(b) Output signal

3.17.5.3.6 Attenuators. One attenuator for each diversity terminal shall be required.

3.17.5.3.6.1 The composite output level of the attenuators shall be discretely adjustable over a minimum range of 0 dB to -35 dB in steps no greater than 1 dB with an accuracy of ±0.5 dB.

3.17.5.3.6.2 The output impedance of the attenuator shall be 600 ohms balanced with the center tap available.

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3.17.5.3.7 Test panel. A test panel shall be provided for each telegraph terminal unless otherwise specified in the contract or order. The test panel shall be mounted in the cabinet in a manner that will permit viewing of the instruments from the front of the cabinet. The panel and instruments shall be completely wired to the necessary points within each terminal so that the operator may perform the desired tests by setting appropriate switches. The panel and switching arrangement shall be designed so that simplicity and ease of operation are optimized. The purpose of the test panel is to provide the instrumentation necessary for test and alignment of the telegraph terminal. Each instrument shall be under switch control so that the operator can electrically place the instrument at the desired points within the terminal. Each instrument shall have two test jacks associated with it so that the instrument may be used for testing at points other than those specified below and one additional test jack for oscilloscope monitoring. Dummy loads shall be supplied where appropriate. The following instruments and corresponding switch selectable test points shall be required.

3.17.5.3.7.1 Instrument - VU meter. A standard volume level indicating meter shall be provided with a step type attenuator to measure levels from -50 to +10 dBm. The attenuator shall be in increments of 2 dB steps. The meter ballistic characteristics shall be that of a standard VU meter. Provision shall be provided for bridging measurements and terminating measurements on the test points shown below. The accuracy of level meter plus attenuator shall be within ±1 dB.

(a) Each output of each keyer
(b) Composite tone out of transmit section
(c) Composite tone in and out of each amplifier
(d) Each input to each convertor

3.17.5.3.7.2 Instrument - Volt/ammeter. A volt/ammeter shall be provided and calibrated in 3 ranges to indicate the 3 modes of input and output telegraph signals, namely (-) 75-0-75, (-) 25-0-25 milliamperes and (-) 10, 0, +10 volts. A screwdriver adjustment shall be provided for zero adjustment. Protection shall be provided to prevent damage to the meter if inadvertently placed in the wrong mode. Meter accuracy shall be 2 percent full scale (F.S).

3.17.5.3.7.2.1 The test point shall be each input and output telegraph loop.

3.17.5.3.7.3 Instrument - delay meter. A differential delay meter calibrated from 0 to 10 milliseconds full scale deflection shall be provided to compare any two channels. A screwdriver adjustment shall be provided for zero adjustment. Meter accuracy shall be 2 percent F.S.

3.17.5.3.7.3.1 Test Points shall be any two channels.

3.17.5.3.7.4 Instrument - dot cycle generator. The dot cycle generator shall generate reversals at 75 and 90 baud with not more than 1 percent total peak telegraphic distortion. The generator shall be switchable to the input of any or all keyers.

3.17.5.3.7.5 Instrument - distortion monitor. A distortion test set shall be provided to work in conjunction with the dot cycle generator for test and alignment. The test panel shall permit the operator to place the dot cycle generator on the input to any or all keyers, connect the output of the transmit section to the input of any convertor and measure the output telegraphic distortion. A suitable visual indication of the distortion shall be required.

3.17.5.3.7.6 Back-to-back. Provision shall be incorporated into the test panel to perform the following functions in any of the subsystem mode of operation.

(a) Back-to-back on an individual channel basis without interfering with the other channels.
(b) Back-to-back of all channels through all modules or components which make up a duplex system.

3.18 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The contractor shall provide and maintain a written quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.2 Quality plan. A quality plan, applicable to the equipments covered by the specification, shall be prepared for Government approval.

4.1.3 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with the inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The contractor shall prepare test procedures for Government approval covering first article and quality conformance inspection.

4.1.5 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
    (3) Environmental inspection
(c) Inspection of preparation for delivery
(d) Reliability testing
(e) Maintainability demonstration test

4.3 First article inspection. Unless otherwise specified (see 6.1.1) one unit shall be required for first article inspection. First article inspection shall consist of examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the inspections specified in Table II.

Table II - First article inspection

Inspection	Reference
Production inspection	4.4.1
Production control inspection	4.4.2
Environmental inspection	4.4.3
Safety factor for overvoltage and overcurrent	3.17.3.1.3 & 3.17.3.1.4

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4.3.1 First article inspection report. A first article inspection report shall be submitted to the Government covering the results of the contractor's first article inspection. (see 6.1.2)

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examination approved in accordance with 4.1.4 and shall include the requirements of Table III.

Table III - Production inspection

Inspection	Reference
Visual	
Keyer nameplate designator	3.17.5.3.1.7
Converter/loop keyer nameplate designator	3.17.5.3.2.5
Safety	3.8
Blown fuse indicators and alarm	3.17.4.1.2
Keyers	
Frequency and frequency shift	3.17.5.3.1
Output level adjustment	3.17.5.3.1.4
Output level	3.17.5.3.1.5
Switches	
Output signal inversion	3.17.5.3.1.3
Control switch	3.17.5.3.1.10
Test monitor jacks	3.17.5.3.1.9
Jitter	3.17.5.3.1.14
Convertor/loop keyer	
Frequency and frequency shift	3.17.5.3.2
Loop keyer ripple voltage	3.17.5.3.2
Switches	
Output signal inversion	3.17.5.3.2.1
Dynamic range	3.17.5.3.2.2
Bias distortion	3.17.5.3.2.3
Telegraphic distortion	3.17.5.3.2.3.1
Time delay adjustment	3.17.5.3.2.4
Test monitor jacks	3.17.5.3.2.7
Diversity combiner	
Combiner operation	3.17.5.3.3
Twinning panels	
Quad diversity operation	3.17.5.3.4
Amplifier	
Frequency response	3.17.5.3.5.1
Impedance, return loss, and longitudinal current suppression	3.17.5.3.5.2 NOTE
Gain variation	3.17.5.3.5.3
Harmonics, hum and noise	3.17.5.3.5.3

NOTE: Impedance shall be measured on one unit. Measurement need not be repeated unless impedance is influenced by an adjustment or if equipment design changes occur.

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Table III - Production inspection (continued)	
Inspection	Reference
Attenuators	
Attenuation	3.17.5.3.6.1
Test panel	
VU meter	3.17.5.3.7.1
Volt/ammeter	3.17.5.3.7.2
Delay meter	3.17.5.3.7.3
Dot cycle generator	3.17.5.3.7.4
Destortion monitor	3.17.5.3.7.5
Back-to-back operation	3.17.5.3.7.6
Voice frequency carrier terminal functional test	4.4.1.1

4.4.1.1 Voice frequency carrier terminal functional test. The terminal shall be connected and operated as specified in 3.17.4.2. The terminal shall also be operated under the following conditions:

(a) Telegraph loops - send and receive

(1) 60 mA, 130 volt neutral (high level)
(2) 20 mA, 60 volt polar (high level)
(3) 1 mA, 6 volt polar (low level)

(b) Operational modes

(1) 4 separate subsystems of 4 channels each
(2) 2 separate subsystems of 4 channels each and one separate subsystem of 8 channels
(3) 2 separate subsystems of 8 channels each
(4) 1 separate subsystem of 12 channels and 1 separate subsystem of 4 channels
(5) 1 separate system of 16 channels

(c) Amplifiers

(1) 16 channel operation (duplex) - 2 on line

(d) Modulation rate - 90 baud

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to Table IV below and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.4. The equipment shall satisfactorily meet the requirements of production control inspection as specified in table IV prior to release for shipment.

Table IV - Production control inspection		
Inspection	Reference	AQL-%
Supply line voltage and frequency	3.10; 3.10.2 & 3.10.3	6.5
Power interruption	3.10	6.5
Electromagnetic interference emission, susceptibility and systems self compatibility	3.16 NOTE 1	6.5
Size	3.17.5	6.5

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Table IV - Production control inspection (continued)		
Inspection	Reference	AQL-%
Convertor/loop keyer		
External battery operation	3.17.5.3.2	6.5
Subsystem operation	3.17.4.1.1	6.5
Impedance, return loss, and longitudinal current suppression	3.17.4.1.1.2 NOTE 2	6.5
Keyer harmonic content	3.17.5.3.1.6	6.5
Frequency accuracy	3.17.5.3.1.13.1	6.5
Input and output characteristics	3.17.3.1.1 and 3.17.3.2.1	6.5

NOTE 1: Electromagentic interference emission, susceptibility, and system self-compatibility shall be performed on one sample unit from the first production control inspection lot. Subsequent testing shall be one unit from each successive 100 units produced.

NOTE 2: Impedance shall be measured on one unit. Measurement need not be repeated unless impedance is influenced by an adjustment or if equipment design changes occur.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in table V and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be approved in accordance with 4.1.4.

Table V - Environmental inspection

Inspection	Reference
Temperature	3.9
Humidity	3.9

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the quality assurance representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.3).

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental inspection may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.4.4 Inspection of preparation for delivery. Inspections shall be conducted to insure conformance with the requirements of Section 5 of this specification.

4.5 Reliability testing. The reliability of the VFCT terminal shall be verified and reported in accordance with MIL-STD-781. The specified MTBF shall be 10,000 hours per channel for reliability testing.

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4.5.1 Demonstration phase of production reliability tests. The demonstration phase of reliability tests shall be conducted with test plan IV at test level A1 in MIL-STD-781. The sample size shall be four production units taken from the initial month's production. Teh reliability requirements shall be successfully demonstrated prior to the shipment of any production equipment.

4.5.2 Production acceptance (sampling) phase of production reliability tests. The production acceptance (sampling) phase of reliability tests shall be conducted with test plan IV-A at test level A1 in MIL-STD-781. The sample size shall be three units from each subsequent month's production. The reliability test shall be successfully completed prior to the shipment of any production lots.

4.5.3 Reliability test plan. The contractor shall prepare a detailed reliability test plan in accordance with MIL-STD-781. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in this specification. Performance parameters to be measured during the reliability test shall be as specified herein. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the Government. No preventative maintenance shall be performed during the reliability test.

4.5.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. Failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service. Only relevant failures shall be counted during the reliability test for an accept or reject decision.

4.5.5 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets the terms and conditions of the contract for acceptance.

4.5.6 Reliability test reports. The contractor shall submit test reports for reliability testing in accordance with MIL-STD-781. Reliability test reports shall be submitted after completion of each reliability test to the command or agency concerned for Government approval (see 6.1.2).

4.6 Maintainability demonstration test. The maintainability requirements of 3.2.3.1 for the equipment shall be successfully demonstrated in accordance with method 3 of MIL-STD-471.

4.6.1 Maintainability demonstration test plan. The contractor shall prepare a detailed maintainability test plan in accordance with MIL-STD-471.

4.6.2 Maintainability demonstration report. A final maintainability demonstration report shall be submitted in accordance with MIL-STD-471 (see 6.1.2).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking as specified in MIL-E-17555 (see 6.1.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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6.1.1 Procurement requirements.

(a) Title, number and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.1.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data require by this specification include, but are not restricted to the following:

(a) Reliability prediction (see 3.2.2.3)
(b) Reliability letter reports (see 3.2.2.5)
(c) Maintainability program plan (see 3.2.3.2)
(d) Maintainability letter report (see 3.2.3.5)
(e) Human engineering plan (see 3.2.4.1)
(f) Equipment requiring human engineering review (see 3.2.4.3)
(g) Human engineering progress reports (see 3.2.4.6)
(h) Change request (see 3.2.4.7)
(i) EMI control plan (see 3.16)
(j) EMI/EMC test plan (see 3.16)
(k) EMC test report (see 3.16)
(l) First article inspection report (see 4.3.1)
(m) Reliability test reports (see 4.5.6)
(n) Maintainability demonstration report (see 4.6.2)

6.2 Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

Preparing activity:
Navy - EC

(Project 5805-N178)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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NAVAL ELECTRONIC SYSTEMS COMMAND
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